

EAE 135, Aerospace Structures

Bending stiffness of a laminated plate or a laminated beam

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Project 2 will require the calculation of the bending stiffness of a laminated beam, which depends on the stiffness of each layer. Due to the layers having different orientations, there is additional work that needs to be done, which requires also knowledge of mechanics of classical lamination plate theory, adapted to composite laminated beams.

We need to know what is the stiffness of each layer of the laminate in direction x_1 (axis of the beam). For 1D bending, the equation of the bending stiffness H_{33}^c for a laminated rectangular cross-section (x_2, x_3) of constant width w is the following:

$$H_{33}^c = \iint E(x_2, x_3) x_2^2 dx_2 dx_3 = \frac{w}{3} \sum_{i=1}^n (E_{x_1 x_1})_i [(x_2)_{i+1}^3 - (x_2)_i^3] \quad (1)$$

For a laminated beam with cylindrical cross-section, the bending stiffness needs to be calculated in the cylindrical coordinate system, and is given by

$$H_{33}^c = \iint E x_2^2 dA = \sum_{i=1}^n \int_0^{2\pi} \int_{r_i}^{r_{i+1}} E (r \sin \theta)^2 r dr d\theta \quad (2)$$

What do you use for $E_{x_1 x_1}$ in Equation 1 or for E in Equation 2?

How do you work through your laminate coordinates?

How do you calculate the axial stresses in each layer?

How do you conduct a failure analysis layer by layer in the laminated beam?

These questions will be answered through this document and the enclosed asynchronous lecture.

The bending stiffness requires the computation of the modulus-weighted centroid (MWC) of the entire cross-section, the definition of the coordinate system with origin in the MWC.

The integral of Equation 2 is converted into a sum, and each layer is accounted for by the coordinates $(x_2)_i, (x_2)_{i+1}$. Each layer is accounted for by the coordinates $(x_2)_i, (x_2)_{i+1}$ with respect to the modulus-weighted centroid, where $i+1$ is for the top of the layer i th.

For a rectangular cross-section that is symmetric, work the calculation from the top half of the mid-plane, top to axis of symmetry, and then double the resulting number to account for the contribution of the lower half-plane.

For this particular laminated cross-section, there is no need to divide the integral into local integrals plus transport terms because the bending stiffness can be computed easily and directly using the cross-section MWC.

Depending on the type of Project 2 assigned, one would use either Equation 1 or Equation 2. This document discusses the calculation of E_{x1x1} for each i th layer of the laminated beam (with a rectangular cross-section or a cylindrical cross-section). It is a reworked version of the relevant paragraph from VLS' notes on Fundamentals of Elasticity.

Fundamentals of Mechanics of Composite Materials

Micromechanics

You will not need micromechanics equations in Project 2 because you will be working with input data provided by us. This section is for completeness.

Micromechanics allows one to calculate the properties of a layer of composite material knowing the properties of the constituents (fibers and matrix) and the fiber volume ratio. Assumptions are that there are no voids, no inclusions, and there is perfect adhesion between fibers and matrix. The composite properties calculated in this way are called 'effective properties'. It is as if one were zooming out of the layer and does not need to see the details, just needs to know that the layer is homogeneous and has a given stiffness, to be used into macromechanics.

The *rule of mixtures* is one of the most famous results of micromechanics: given a fiber with Young's modulus E_{11f} , a matrix material with Young's modulus E_m , and the fiber volume ratio V_f (amount of fiber volume with respect to the volume of the composite), the composite Young's modulus E_{11c} in the fiber direction is given by:

$$E_{11c} = E_{11f}V_f + E_m(1 - V_f)$$

This relation has been shown experimentally to be accurate for the composite longitudinal Young's modulus E_{11c} (in the fiber direction) and the Poisson's ratio ν_{12c} in the composite plane:

$$\nu_{12c} = \nu_{12f}V_f + \nu_m(1 - V_f)$$

The following two equations on respectively the transverse modulus E_{22c} and the in-plane shear modulus G_{12c} , on the other hand, are not very accurate, but may provide ballpark values.

$$\frac{1}{E_{22c}} = \frac{1}{E_{22f}}V_f + \frac{1}{E_m}(1 - V_f)$$

$$\frac{1}{G_{12c}} = \frac{1}{G_{12f}}V_f + \frac{1}{G_m}(1 - V_f)$$

Macromechanics

Macromechanics looks at an entire layer as part of a laminate. For a laminate, we could also call this as "classical lamination plate theory" or "classical lamination beam theory".

This theory is used to calculate the stress/strain of a layer of composite, knowing the effective properties, and then the layers make a plate or a beam.

The layers can then be stacked on each other in a given stacking sequence, and knowing the forces and moments applied to this plate (input data), one can calculate the strain of the plate and then the stress in each layer, and finally whether failure occurs, and which layer is going to fail first.

For composite plates or beams (rectangular and cylindrical cross-sections), it is often possible to make the assumption of plane stress: in a plate defined by a global coordinate system, if the thickness is much smaller than the other dimensions of the plate, and there is no out-of-plane loading, then those stress components are zero. The same holds in the reference system where 1 is the direction of the reinforcement, 2 is the in-plane transverse direction (perpendicular to the reinforcement): $\sigma_{33} = \sigma_{23} = \sigma_{13} = 0$. Note that in EAE 135, we use a global reference system (x_1, x_2, x_3).

For composite laminates, we consider a material that is linear elastic material and has 9 independent constants. It is called an orthotropic material. The constitutive equations in the local reference plane of the lamina are shown below in **Equation 3**:

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ & C_{22} & C_{23} & 0 & 0 & 0 \\ & & C_{33} & 0 & 0 & 0 \\ & & & C_{44} & 0 & 0 \\ & & & & C_{55} & 0 \\ sym & & & & & C_{66} \end{bmatrix} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{Bmatrix} \quad (3)$$

The thin layer (or lamina) is unable to carry stresses $\sigma_{33}, \sigma_{23}, \sigma_{13}$. **Equation 3** is simplified into the following equation (constitutive equation for an orthotropic lamina under plane stress):

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{Bmatrix} = [Q] \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{Bmatrix} \quad (4)$$

where Q is given by

$$Q = \begin{bmatrix} \frac{E_{11}}{1 - \nu_{12}\nu_{21}} & \frac{\nu_{21}E_{11}}{1 - \nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{12}E_{22}}{1 - \nu_{12}\nu_{21}} & \frac{E_{22}}{1 - \nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix} \quad (5)$$

where $\nu_{21} = \nu_{12} E_{22}/E_{11}$. ν_{21} is the minor Poisson's ratio.

Direction 1 is the direction of the fibers in a lamina, direction 2 is the in-plane direction perpendicular to direction 1. Direction 3 is the out-of-plane direction.

[Q] is the stiffness matrix of an orthotropic lamina under plane stress, in the material reference

system (1,2), and is symmetric. It is also called **reduced stiffness matrix in the local reference system**. Note that the normal stress and the shear stress are uncoupled with respect to each other ($Q(1,3) = Q(2,3) = 0$).

What happens if one wants to form a plate (or laminate) with different layers (also called laminae or plies), at different orientations? Most aerospace-grade composite structures are made by more than one single layer of material, to cover multiple loading directions. To design a laminate, one needs to use a different reference system, called global (in plane) system, which is indicated as (x, y) in **Figure 1**.

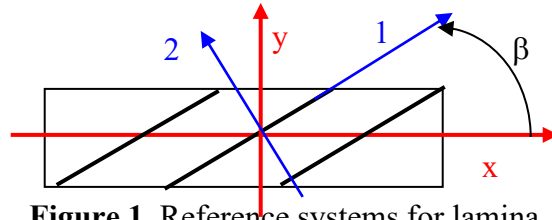


Figure 1. Reference systems for lamina.

β is the angle between axis x of the global reference system, and axis 1 of the material reference system, aligned with the reinforcement. A transformation is required from one system to the other. In western literature, the stiffness matrix of a layer in the global reference system is called $\bar{Q}$ (also labeled $\bar{Q}$ below).

$\bar{Q}$ is a symmetric 3x3 matrix that is typically fully populated, i.e. $\bar{Q}_{13}$ and $\bar{Q}_{23}$ are different from zero, unless β assumes particular values (0 deg. and 90 deg. will have those terms = 0). The presence of non-zero $\bar{Q}_{13}$ and $\bar{Q}_{23}$ means that there is coupling normal stress/shear, which complicates the design/analysis and failure prevention problem.

The easiest way to calculate $\bar{Q}$ is the following:

$$\bar{Q} = T^{-1} \cdot Q \cdot R \cdot T \cdot R^{-1}$$

$$\text{where } T = \begin{bmatrix} \cos^2 \beta & \sin^2 \beta & 2 \sin \beta \cos \beta \\ \sin^2 \beta & \cos^2 \beta & -2 \sin \beta \cos \beta \\ -\sin \beta \cos \beta & \sin \beta \cos \beta & \cos^2 \beta - \sin^2 \beta \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad (6)$$

As each layer can be characterized by its orientation and stiffness in the global reference system, the layers are bonded together to form a laminate, which will have a stiffness matrix depending on the $\bar{Q}$ matrices of each layer, as well as on the position of the layer in the plate.

Design and analysis of the composite plate can now be done, and failure of each ply (as well as the failure sequence, called ply-by-ply failure) can be assessed. The failure analysis will require the analysis of the loads applied globally on the entire structure, and the analysis of how that stress is imparted to the individual layers, and how it compares to the individual layers' strengths (as in, tensile longitudinal (or axial), compressive longitudinal (or axial), tensile transverse, compressive transverse and in-plane shear strengths).

Back to the Bending Stiffness

In this course, we use Bauchau & Craig nomenclature, based on $x = x_1$ being the axis of the beam, and the cross-section having coordinate $y = x_3$. In bending problems, the key variable is x_1 . Take a piece of paper, mark vertical lines along the longest axis of the paper: "1" is the direction of the lines, "2" is the direction of the shortest axis of the piece of paper. Now, each layer could have a different orientation. For a 0 deg. layer, axis "1" coincides with axis "x1".

If the plies have different orientation, we will need to find out what is the stiffness in direction x_1 for each layer. We will need to invert the $\bar{Q}$, obtaining the so-called reduced compliance matrix in the global coordinate system, $\bar{S}$. The axial stiffness of each layer is then

$(E_{x1x1})_{ith} = \frac{1}{(\bar{S}_{11})_{ith}}$, which is used in the summation term of Equation 1 or 2 (depending on the type of the assigned Project). We have not finished the process yet, because we still need to compute the axial stress and

Steps for EAE 135 Project 2

These are the steps required for a typical Project 2. Project 2 changes from year to year and these steps are meant to guide you through the process. We generally have a laminated beam (either rectangular or cylindrical).

- 1) Leave H_{33}^c symbolic as you integrate your governing differential equation. Your bending stiffness is constant along the entire spar, we never give taper.
- 2) You will need to code this part because you are iterating over the TBD layers. If you have a rectangular cross-section, work on the the top flange, get half of your bending stiffness, and remember to double it at the end. If you have a cylindrical cross-section, you will get your entire bending stiffness as you work through the integral in Equation 2.
- 3) Compute the Q matrix using the material properties of the CFRP composite from the table provided in the Canvas folder. Pay attention to the calculation of minor Poisson's ratio. The Q matrix is the same for all the layers.
- 4) Calculate Qbar doing an iteration on the orientation angle. If you use MATLAB, remember to use the correct trigonometric functions (*sin*, *cos* need radians). If your calculations are correct, this is what you would find:

$\bar{Q}(0 \text{ deg.}) = Q$; $\bar{Q}(1,1)$ for the 90 deg. = $Q(2,2)$; $\bar{Q}(1,1)$ for the +/- 45 deg. = $\bar{Q}(2,2)$ for those orientations; $\bar{Q}(3,3)$ for the +/-45deg. is higher than $Q(3,3)$.

Each of these Qbars needs to be inverted.

$\bar{S}(\text{for a given angle}) = \text{inv}(\bar{Q}(\text{for that angle}))$

4) Extract value $S_{bar}(1,1)$. $(E_{x1x1})_{ith} = \frac{1}{(\bar{S}_{11})_{ith}}$. This value of the projected Young's modulus for each layer will be used in Equation 1.

5) Now you go to Equation 1 of this document if you have a rectangular cross-section, or to Equation 2 if you have a cylindrical cross-section.

Use the coordinates of each layer with respect to the modulus-weighted centroid, which is the center of the cross-section (due to geometric and material symmetry). For a cylindrical cross-section, if you roll a sheet a paper into a cylinder and then you unroll it, you have a plate, and the equations of classical lamination plate theory hold.

Assign coordinates $i+1$ to the top of the layer, i to the bottom of the layer, for each layer. If you have a core material, its contribution in stiffness can be approximated to be zero with respect to the carbon composite. The core material's role is to provide a significant bending stiffness by moving the stiffer CFRP layers away from the midplane, maximizing the coordinate x_2 (or r , if you are using cylinders).

6) Once you get this operation completed, you can replace this value into the solution of the governing equation for the axial displacement as a function of x_1 , and the solution for the bending moment M_3 as a function of x_1 .

6) The axial stress analysis and the following failure analysis through the thickness will be conducted where at the value x_1 where the axial stress is the highest; if under bending and in case of a cantilever beam, the root is the location with the highest bending moment. When you calculate the axial stress in a rectangular cross-section, use

$$\sigma_1(0, x_2, x_3) = - \frac{(E_{x1x1})_{ith} \cdot x_2 \cdot M_3(0)}{H_{33}^c}$$

If you have a cylindrical cross-section, you can use the equation as well for cantilever boundary conditions (draw your cylindrical cross-section; x_2 is the vertical axis of the cross-section).

Remember to use the bending stiffness **of the entire cross-section** in the denominator. For simplicity, use the coordinate x_{2i} and x_{2i+1} for respectively bottom and top of each layer. You can also select an x_2 in the center of the layer, but make sure to use the correct stiffness. Once you calculate the top axial stress for each layer, give the opposite value to the corresponding layer in the bottom half plane and finish this part of the stress analysis.

Finally, you need to find out if this spar cross-section is failing, doing a layer-by-layer failure analysis. Use the following properties from the tables_composites.pdf file (pages 2-3):

axial tensile strength for 0 deg. = axial tensile strength X_T

axial compressive strength for 0 deg. = axial tensile strength X_C . (last table of the file tables_composites.pdf, remember to convert from ksi into MPa)

axial tensile strength for 90 deg. = transverse tensile strength Y_T

axial compressive strength for 90 deg. = transverse compressive strength Y_C (last table of the file tables_composites.pdf, remember to convert from ksi into MPa)

axial tensile strength for +45, -45 deg. = transverse tensile strength Y_T

axial compressive strength for +45, -45 deg. = transverse compressive strength Y_C (last table of the file tables_composites.pdf, remember to convert from ksi into MPa)

Incorporate 1.5 or a 1.25 factor of safety for respectively airworthy and spaceworthy structures.